

Welcome!

DSC 340 Applied Machine Learning

5 February 2025

About this Course. What is Data Science? The Data Science Lifecycle.

The Plan

- Welcome!
- Introductions
- Course Overview
- What is Data Science?
- What is Applied Machine Learning?
- Data Science Lifecycle (first look)

Next 2 days: Example data analytics problem (linear regression)

About DSC 340

Syllabus

About the Course (Note: formerly DSC 305)

- Tools
- Workflow (Data Analytics Lifecycle)
- Assess
- Assess and draw conclusions
- Communicate
 - Orally
 - In writing
 - Visualization
- Hands-on experience with varied datasets

Before this course

Prerequisites:

- DSC 230 Statistics foundation, linear regression
 - MAT 130 Intro stats
- DSC 270 Python, NumPy, Pandas, SQL, Excel, etc.
 - CSC 170 Intro coding
- CSC 165 Introductory calculus (minimizing a function)
- Or see instructor ASAP!

After this Course

Learning outcomes:

- What a data scientist does
- Apply data analytics lifecycle to a problem
- Tools: Py3, Jupyter, pandas, sklearn, Matplotlib, tf? ...
- Jupyter = Code + Output + Markdown (browser)
- Data sets

After this Course (cont.)

- Exploratory data analysis
- Prep data for model building
- Methods (e.g., logistic regression)
- Model data appropriately (e.g., scale)
- Occam's razor
- Analyze and assess results
- Present results orally and writing (+ data viz)
- Experience: labs + significant data analytics project

Resources

- Textbooks: EMC, PDSH
- Many (!) online resources
- Jupyter Notebook (Olin 208, Anaconda, CoLab)
- Office Hours: Weekdays at 3:00 PM (for now)
- Disabilities, Library, ESL, Computer Labs (Olin 208, etc.)
- Coding Collaborators (Partners): P2-P3 and P4-P5

Grades

• Data Analytics Projects ("Labs") (about 5 @ 5%)	25%
• Final Project	8%
• Tests (about 6, usu. on Wednesdays)	54%
• Comprehensive Final Exam	13%

	100%

Workflow and Late Submissions

- Attendance
- No retakes*
- No late submissions*
- Academic honesty
- Class participation
- Respect for others
- Email, Moodle

*See syllabus for discussion of special cases

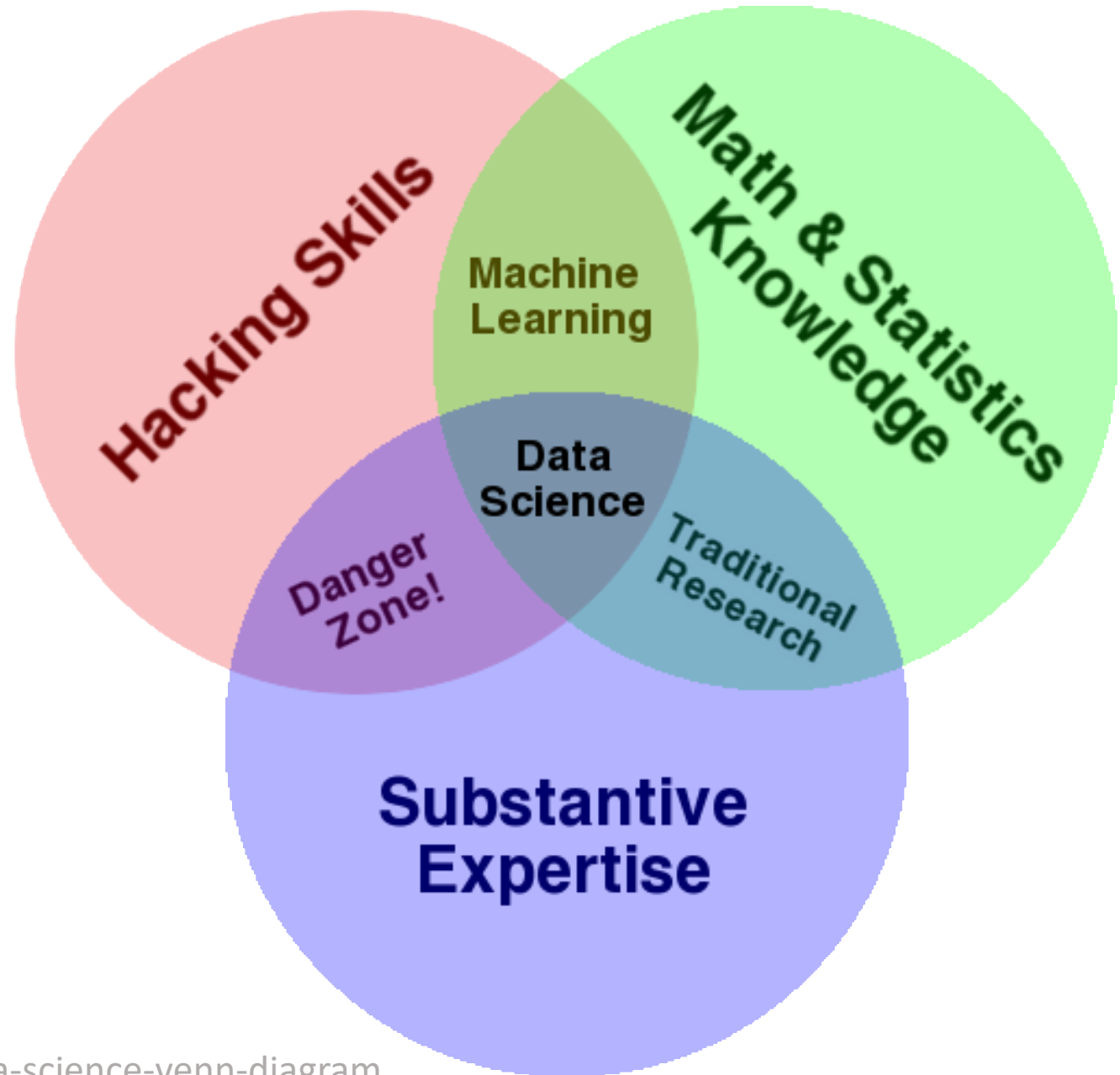
Questions?

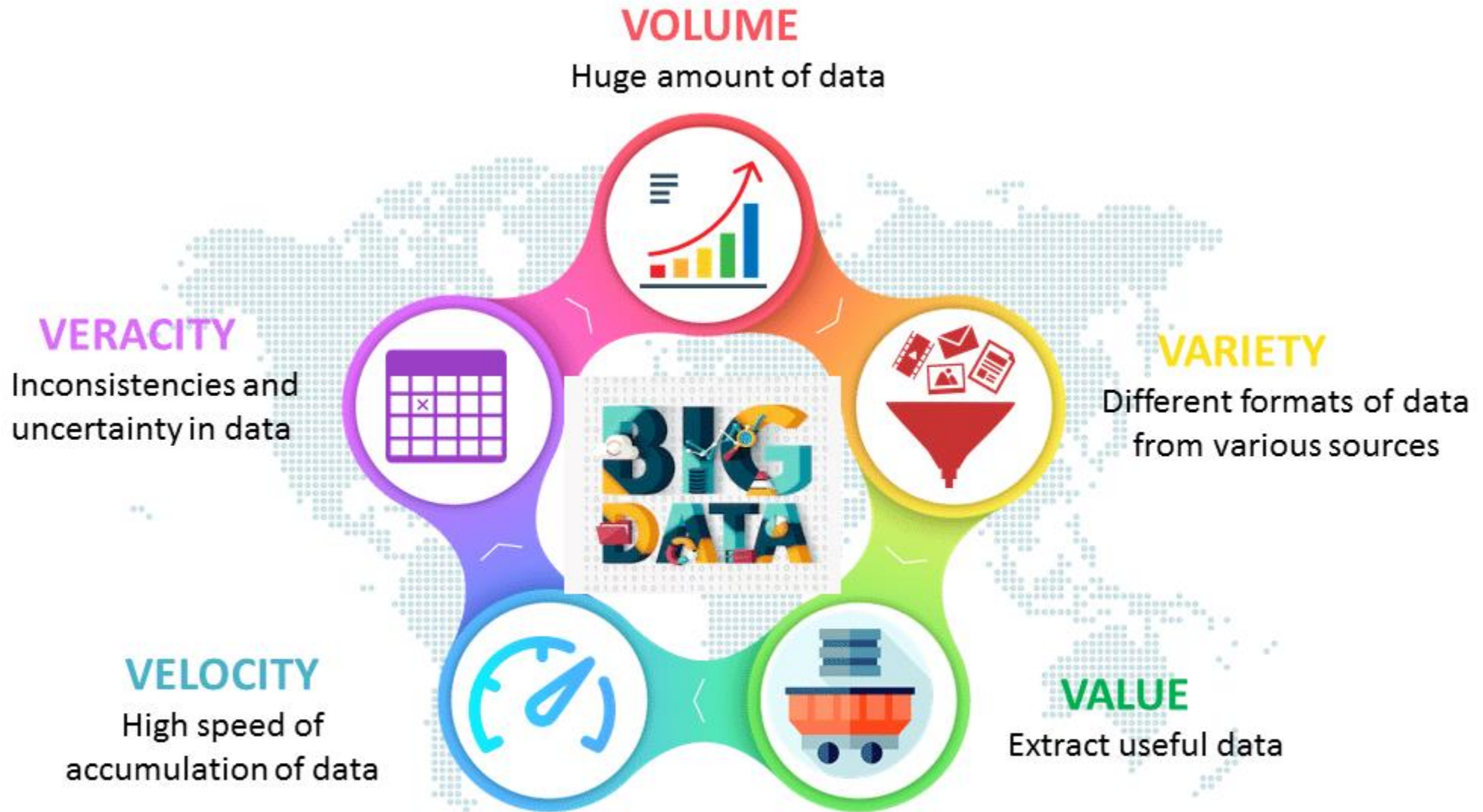
- What are your questions about this course?

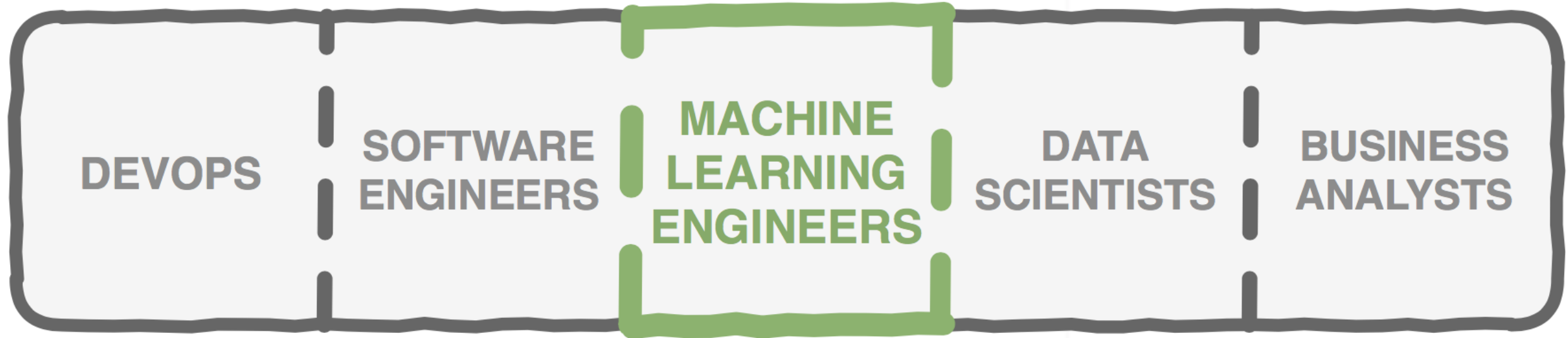
What is Data Science?

EMC 1, PDSH Preface, Other Resources (see Moodle)

Data Science







<https://medium.com/@tomaszdudek/but-what-is-this-machine-learning-engineer-actually-doing-18464d5c699>

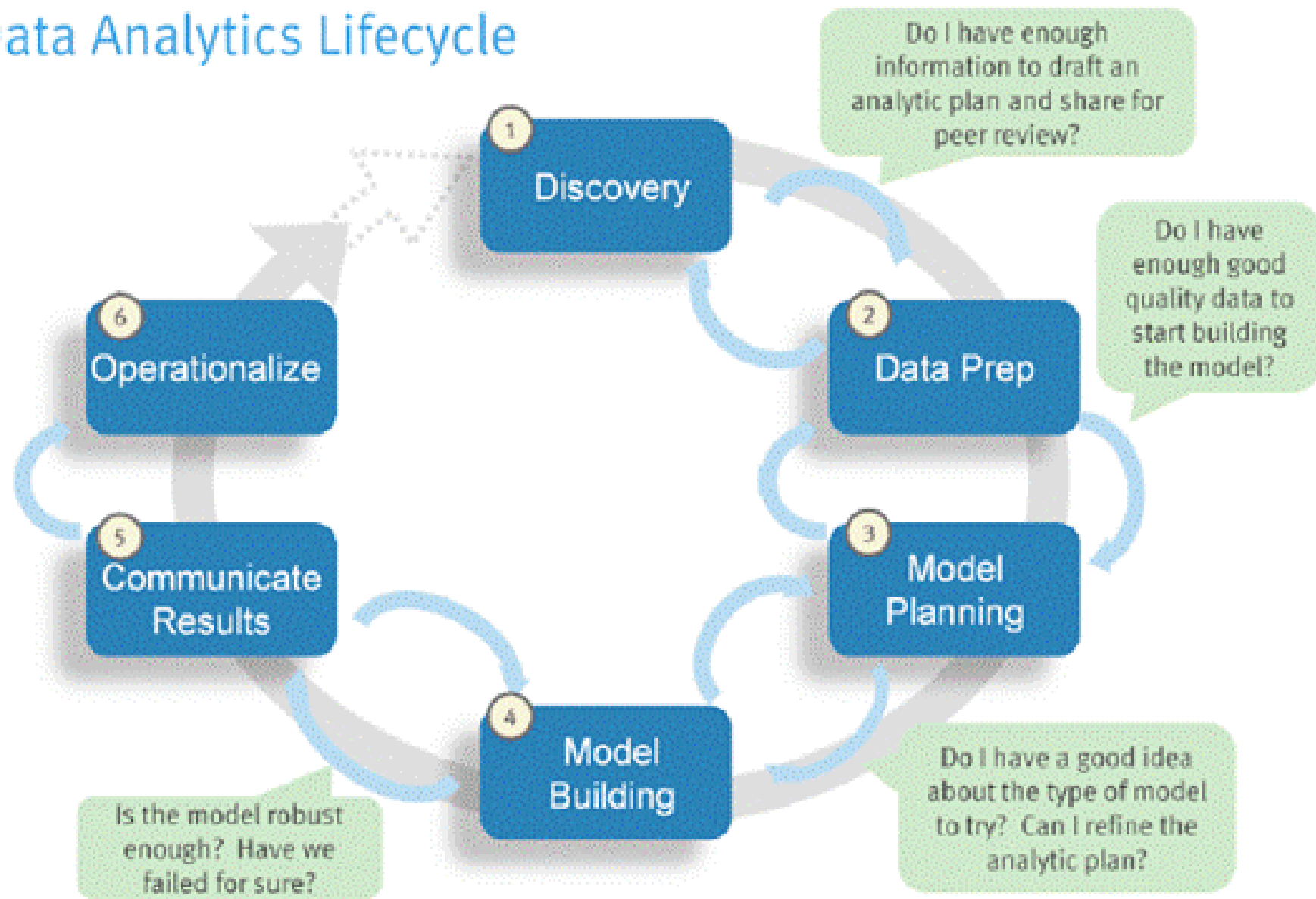
Data Analytics Roles

- **Organization User:** understands domain area, benefits from work
- **Project Sponsor:** high-level direction from administration
- **Project Manager:** supervises process to meet milestones/objectives
- **BI Analyst:** Business domain expertise, dashboards and reports
- **DB Administrator:** IT support for DB: roles, permissions, installations
- **Data Engineer:** Programming, complex SQL queries,
- **Data Scientist:** The expert in the room for techniques, models, and evaluating result: Choose the right method for the problem at hand.

Data Analytics Lifecycle

EMC 2

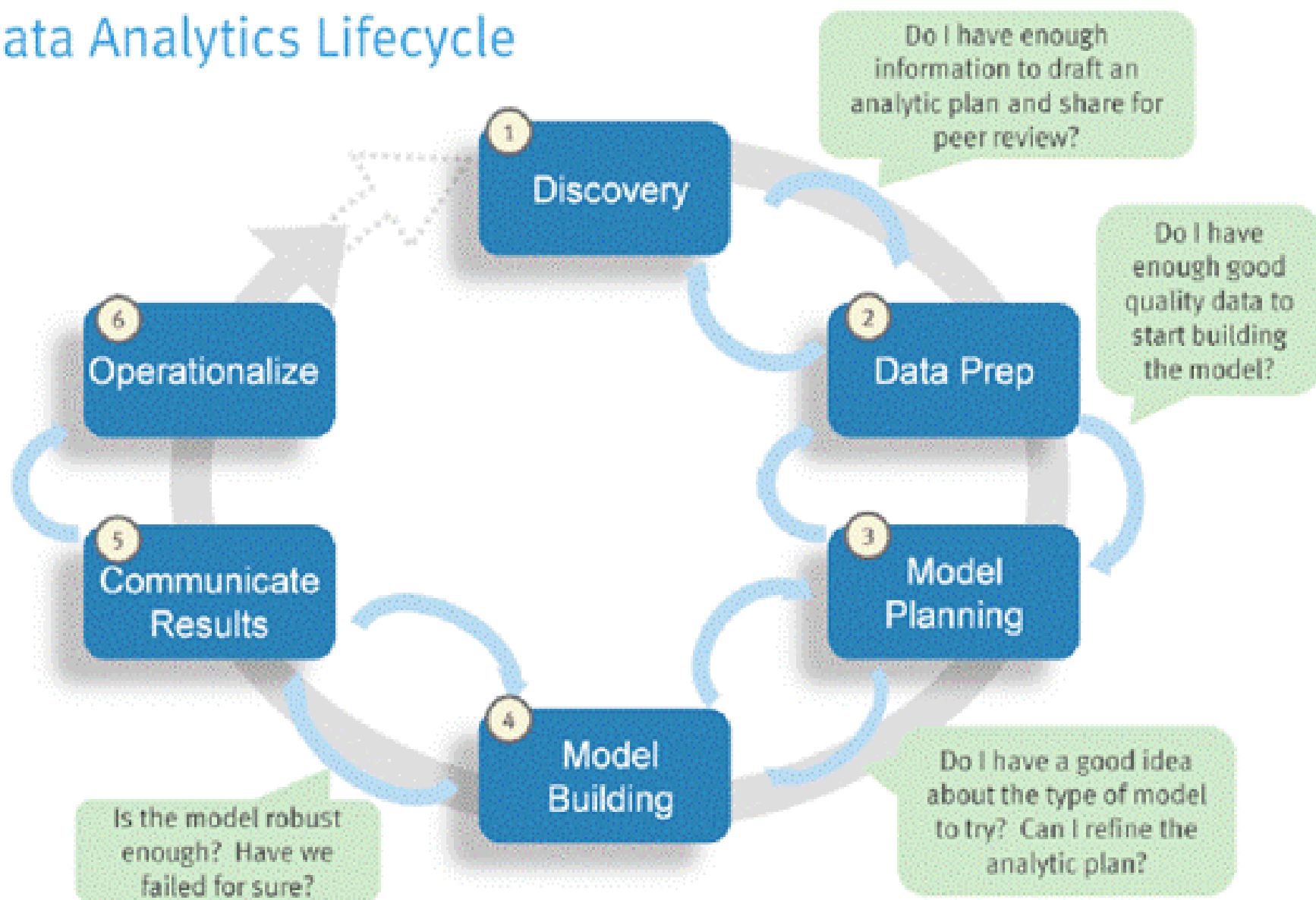
Data Analytics Lifecycle



1. Discovery

- Learn the domain (business, science, etc.)
- What resources do we have? (people, tech, tools, data?)
- What problem are we solving? (frame the problem)
- Who are the stakeholders? (Who is affected? What are the risks?)
- Interview the Analytics Sponsor
- Developing initial hypotheses
- Identifying potential data sources

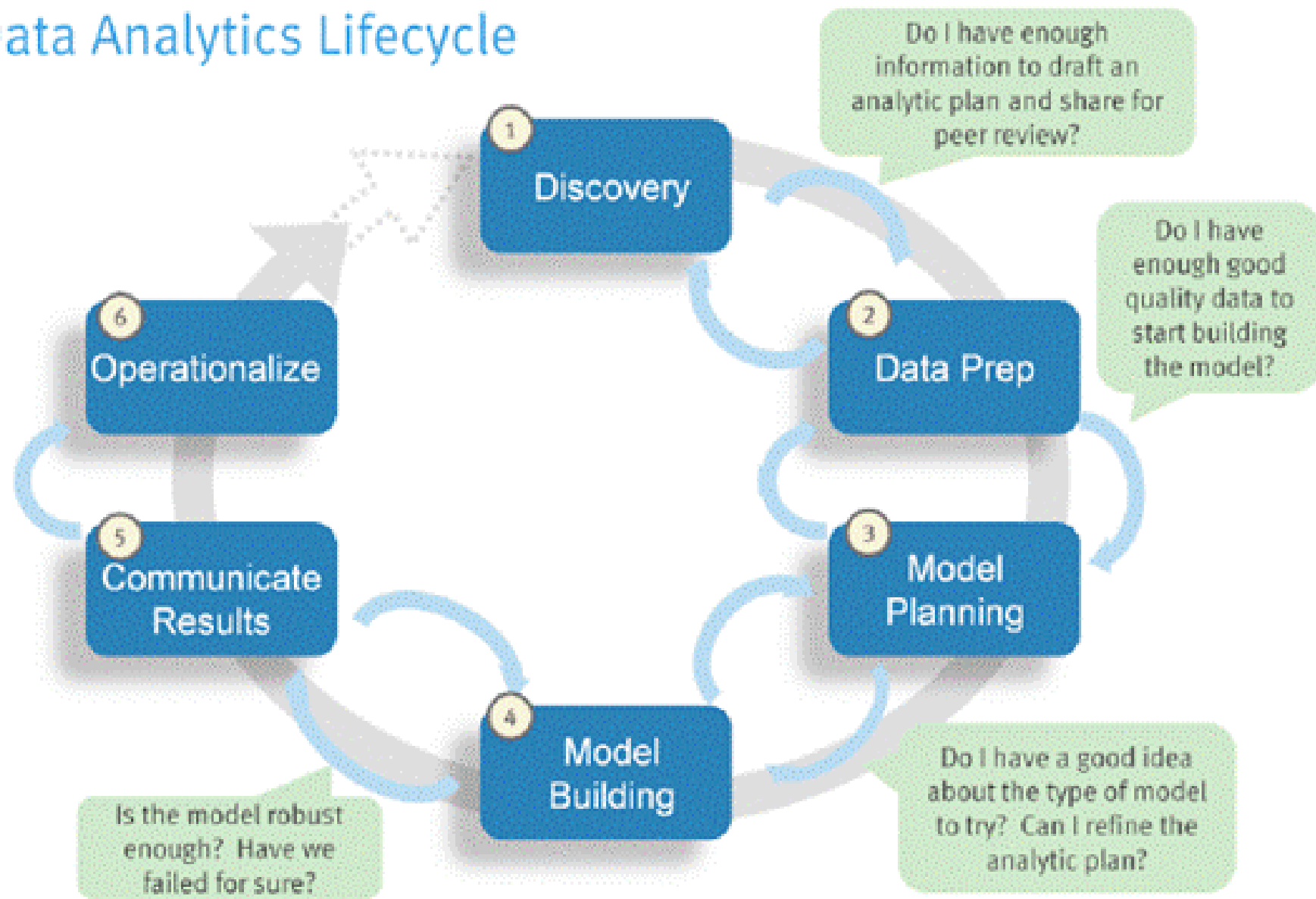
Data Analytics Lifecycle



2. Data Preparation

- Sandbox data (no live databases!) analytic vs. operational DB
- Make sure you have the resources (data, tech) you need
- Learn about the data (semantics, missing values, errors, duplicates ...)
- Data conditioning (address issues discovered, structure)
- Survey and visualize (is it representative? Consistent? Time frames?)

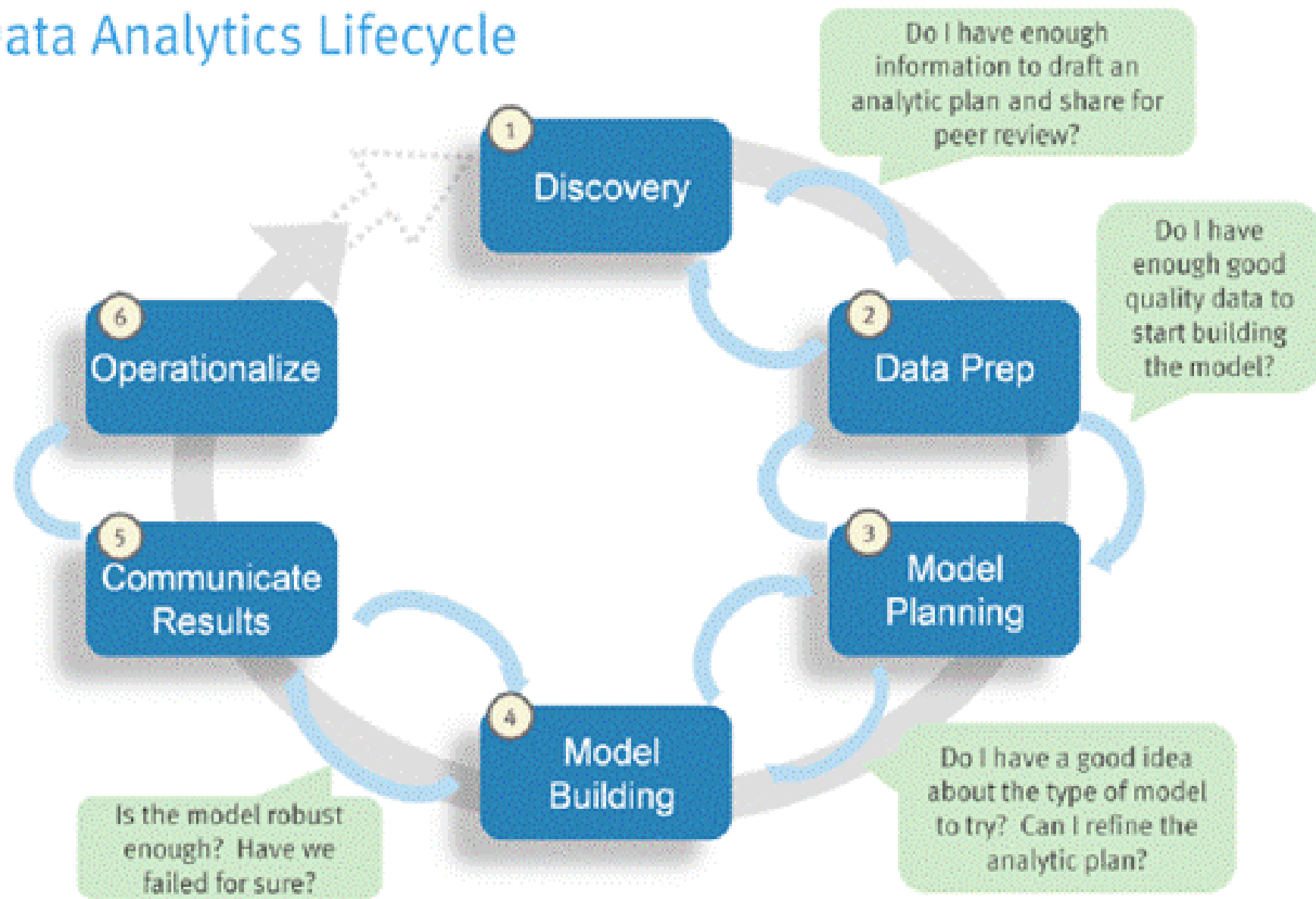
Data Analytics Lifecycle



3. Model Planning

- Data exploration and variable selection
 - Already cleaned, but need to understand relationship between variables
- Model selection:
 - Is this a classification problem? Clustering?
 - Which method(s) are most appropriate?

Data Analytics Lifecycle



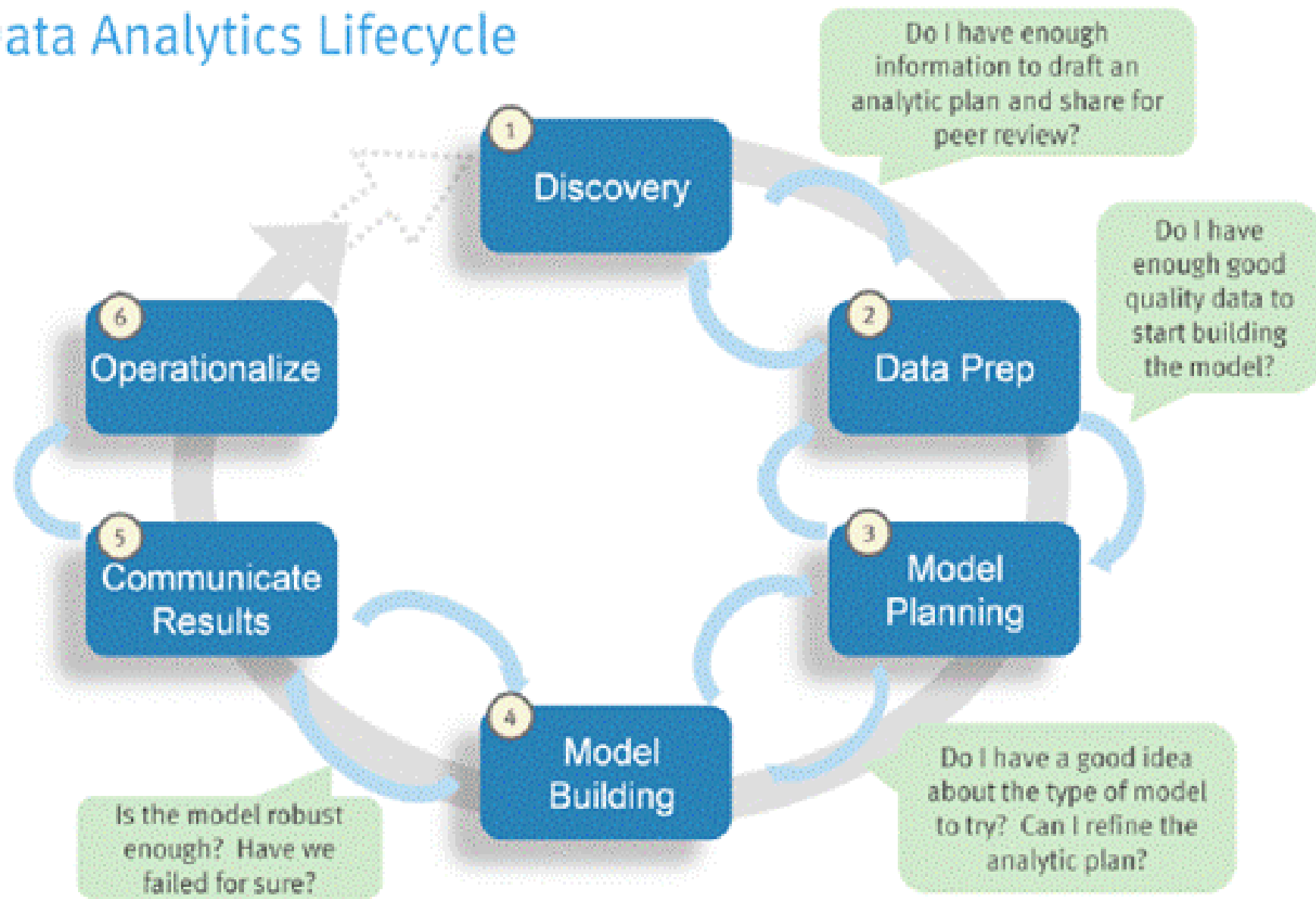
4. Model Building

- "Train" model ("fit" data)
- Does the model appear valid and accurate?
- Does the output/behavior make sense to domain experts?
- Do the parameter values of the fitted model make sense in domain?
- Is the model sufficiently accurate to meet the goal?
- Does it avoid intolerable mistakes? (depends on context!)
 - For example: false positives. How bad?

4. Model Building (cont.)

- Do we need more data? Transformations? Less data?
- Will the model support runtime (operational) requirements?
- Is a different form of the model needed for operationalization?

Data Analytics Lifecycle



5. Communicate Results

- Did we succeed or fail?
- Common sense evaluation
- Present key results
- Make recommendations for follow-up
- Document work of team

Data Analytics Lifecycle

